Venkata Sai

Software Engineer

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Professional Experience

Senior Software Engineer *Jul 2025 – Present*

EPAM Systems | Large-Scale AI Training Infrastructure

- Designed and implemented core components for **OpenAI API** integration using **Python** and **FastAPI**, facilitating seamless **AI systems** deployment across various research projects, enhancing workflow efficiency by 30%.
- Led optimization of **CI/CD** pipelines using **Jenkins** and **Kubernetes** for **AI/ML models**, reducing build and deployment times by 45% and improving overall developer experience significantly.
- Developed an internal **developer experience** portal with **Python Django** and **React**, providing self-service tools for experiment management, decreasing setup time for new engineers by 25%.
- Implemented a robust **testing strategy** for **TensorFlow** and **PyTorch** models within **AWS EKS**, ensuring high reliability and data integrity, catching 90% of regressions pre-deployment.
- Engineered **agentic AI workflows** leveraging **LangGraph** and **OpenAI API** for automated code review suggestions, streamlining development cycles and boosting team productivity by 20%.
- Automated **model deployment** with **Terraform** on **AWS Lambda** and **S3**, creating scalable **AI inference** endpoints, reducing manual intervention by 60% for frequent updates.
- Contributed to enhancing **training frameworks** with **Python** and **Keras**, developing reusable **libraries and systems** that improved model convergence speed by 10% for research teams.

Software Engineer *Jul 2022 – Jun 2025*

Globant | Enterprise AI/ML Solution Development

- Developed high-performance **Python FastAPI** services to manage **AI inference** requests, integrating with **OpenAI API** for various client applications, processing 50K requests/minute.
- Optimized data **workflows** for **ML training** datasets using **Apache Airflow** and **AWS S3**, improving data ingestion reliability and reducing processing time by 30% for analysts.
- Implemented a comprehensive **testing strategy** for backend **Python** services, using **Pytest** and **Docker**, significantly enhancing code quality and reducing post-release bugs by 40%.
- Improved existing **CI/CD** pipelines in **GitLab CI/CD** for faster code deployment to **AWS EC2** instances, reducing release cycles from bi-weekly to weekly for the entire team.
- Collaborated on designing microservices architecture for new features using **Python** and **Flask**, enabling independent development and deployment, decreasing feature delivery time by 20%.
- Developed **monitoring** dashboards using **Datadog** for **Python** applications, providing real-time insights into system health and performance, decreasing incident resolution time by 25%.
- Automated infrastructure provisioning with **Terraform** on **Azure** for various project environments, standardizing setups and reducing manual configuration effort by 50% across teams.
- Contributed to **developer experience** improvements by refactoring legacy **Python** codebases, enhancing readability and maintainability, reducing onboarding time for new engineers by 15%.

Software Engineer *Jul 2020 – Aug 2022*

Nagarro | Customer Engagement Platform

- Developed robust **RESTful APIs** using **Python Django** for a customer engagement platform, handling sensitive user data securely and supporting 1M daily user interactions.
- Implemented core backend logic for new features using **Python** and **PostgreSQL**, ensuring data integrity and optimizing query performance, improving response times by 20%.
- Configured and maintained **CI/CD** pipelines using **Jenkins** for automated build and deployment processes, reducing manual errors by 70% and accelerating software delivery.
- Built microservices using **Python Flask** to decouple monolithic components, enhancing system maintainability and enabling independent team deployments, reducing downtime by 10%.
- Authored comprehensive unit and integration tests using **Pytest** for critical application modules, increasing code coverage to 85% and significantly boosting application reliability.

Skills

Languages: Python
Frameworks & Tools: FastAPI, Django, Flask, Keras, PyTorch, TensorFlow, Docker, Kubernetes, Jenkins, Terraform, Pytest, Apache Airflow, Datadog, React, GitLab CI/CD, OpenAI API, LangGraph
Cloud & Infra: AWS S3, AWS Lambda, AWS EKS, AWS EC2, Azure, PostgreSQL
Concepts: RESTful APIs, Microservices, CI/CD, Scalability, Distributed Systems, System Design, Agile Methodologies, Monitoring, Testing Strategy, Developer Experience, Workflow Optimization, Automation, AI Systems, Training Frameworks, Model Deployment, Libraries and Systems

Education

Master's in Computer Science, Oklahoma State University, Stillwater
Bachelor's in Computer Science, JNTU Hyderabad, India